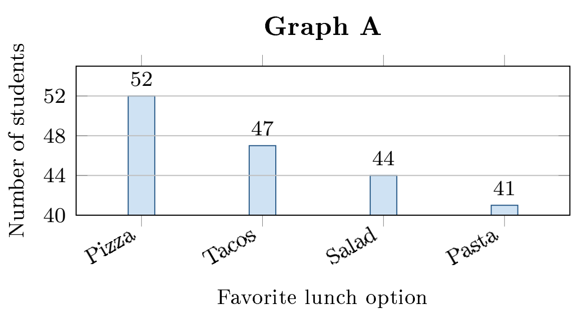
NAME: ______________________________ **PERIOD:** ________

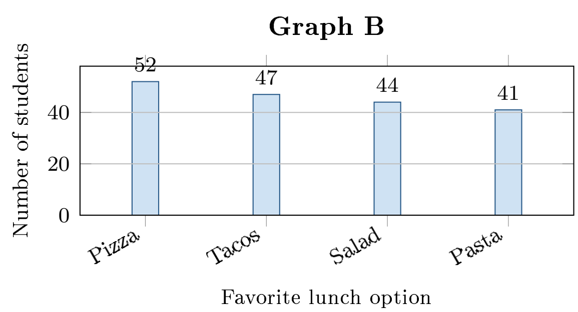
Same Counts, Different Message

AP Statistics · Unit 1 · Topic 1.4

[STAR] THE PROBLEM

Suppose 184 students chose a favorite lunch option. A newsletter printed **Graph A** with the headline "**Pizza wins by a landslide.**" A council member redrew the data as **Graph B** and said, "It is basically a tie."





FIRST TAKE — one sentence before you work the parts

Is pizza clearly the favorite? Name the graph that most influenced you.

[BOOK] GRAPHING ONE CATEGORICAL VARIABLE (keep on the page)

Bar height represents a category's count or relative frequency. Label both axes, use equal-width bars, and start the vertical scale at zero to avoid distortion. The visible height above a displayed baseline is the count minus that baseline. Compare what the lengths suggest with the actual counts before writing a claim.

Work (a)–(c).

DOK 1 **(a)** Read the Pizza and Pasta counts. How many more students chose Pizza?

DOK 2 **(b)** Graph A starts at 40. Find the visible heights above 40 for Pizza and Pasta. Compare those heights with the actual counts.

DOK 3 **(c)** Which graph gives a fairer picture of the lunch votes? Explain how its scale affects the comparison. Use your counts to judge both 'landslide' and 'basically a tie,' then write a more accurate headline.

> **Frame:** Graph _____ is fairer because its scale ______________________________. Graph A makes ________________________ look _______________.

> **Frame:** The actual counts are ____________________, so my headline is: ____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.